Consumer Complaint Classification under Small-Data Constraints

Mini-Project Presentation

Khayrislamov Timur Petrenko Maksim

Natural Language Processing (NLP) in Python

Research Question and Task

- **Goal:** predict the *product category* from each consumer complaint narrative.
- **Setting:** balanced 5-class classification problem under **small-data constraints**.
- **Questions:**
  - How does performance change as the number of labeled examples per class increases?
  - Are pretrained embeddings better than locally trained embeddings in the low-data regime?
- **Classes:** Bank account or service, Credit card, Credit reporting, Debt collection, Mortgage.

Models and Experimental Setup

Models

- TF–IDF + Logistic Regression
- Local Word2Vec + MLP
- Pretrained GloVe + MLP

Evaluation

- Main metric: Macro F1
- Accuracy also reported
- Balanced subset sizes: 100, 300, 600, 1000, 2000 per class
- 3 random seeds per size
- Stratified train/validation/test split

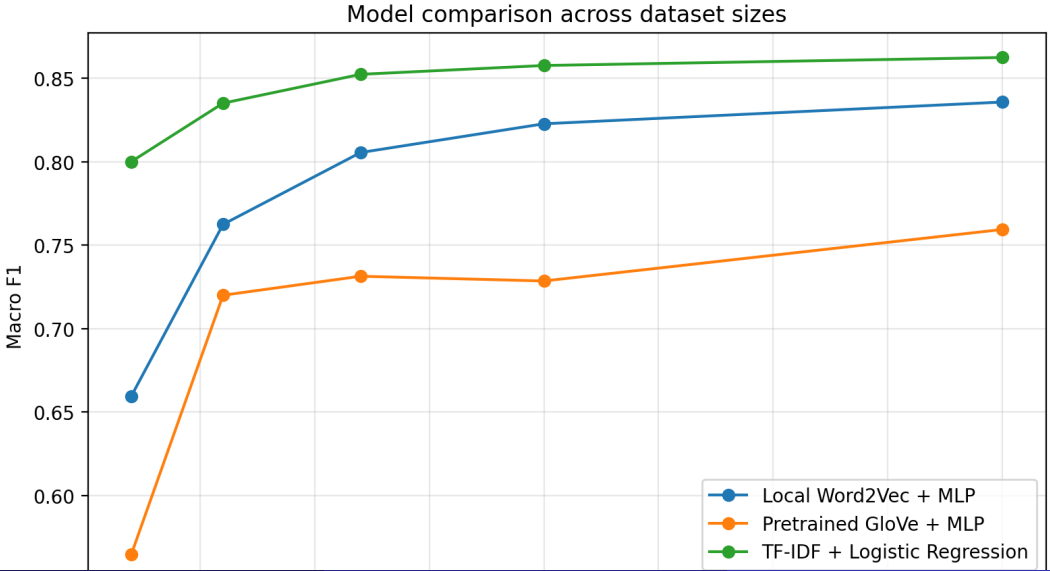
Motivation: compare a strong classical sparse baseline against embedding-based document representations.

Single Full-Data Run

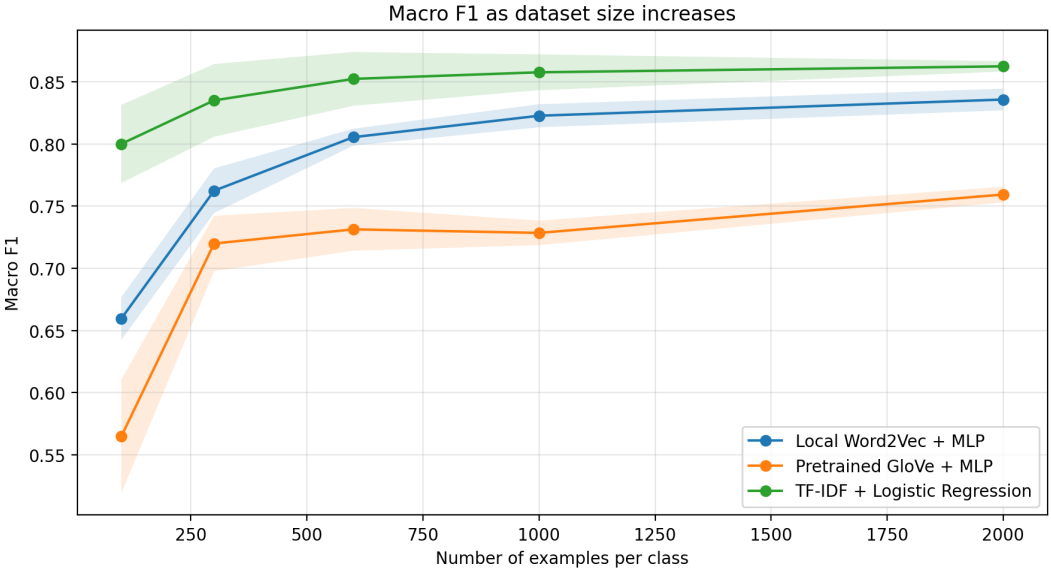
Model	Accuracy	Macro F1
TF–IDF + Logistic Regression	0.8679	0.8680
Local Word2Vec + MLP	0.8435	0.8423
Pretrained GloVe + MLP	0.7454	0.7427

- TF–IDF is the strongest overall model.
- Local Word2Vec is competitive, but still below the sparse baseline.
- Pretrained GloVe is clearly the weakest approach.

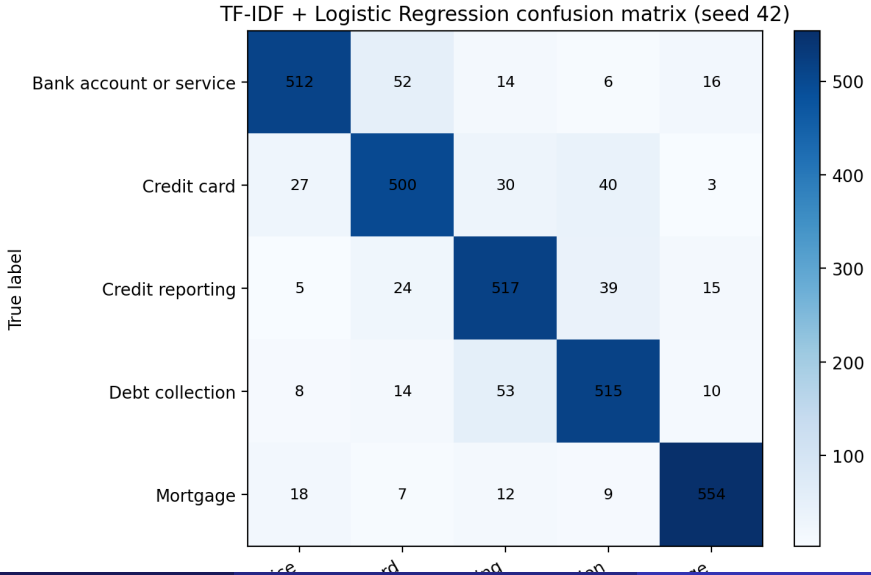
Main Result: Macro F1 Across Dataset Sizes



Repeated Study Curves with Variability



Error Analysis of the Best Model



Why These Errors Happen

- **Debt collection vs Credit reporting:** many complaints discuss collection accounts through the lens of credit-report damage, so both label vocabularies appear together.
- **Bank account vs Credit card:** narratives about payments, fees, credit lines, and bank-issued cards blur the boundary.
- **Mortgage vs Bank account:** escrow, autopay, and home-equity complaints sometimes use generic banking language.

Interpretation: the strongest remaining mistakes are mostly boundary cases, not random failures.

Conclusion

- TF–IDF + Logistic Regression outperforms both embedding-based models at every dataset size.
- Local Word2Vec consistently beats Pretrained GloVe, showing the value of **domain-specific representations**.
- The task is highly **lexicalized**: sparse features with bigrams are extremely effective.
- Repeated runs and the error analysis make the conclusions more credible.

Main takeaway:

Simple classical baselines remain very strong on applied NLP tasks with rich lexical cues.